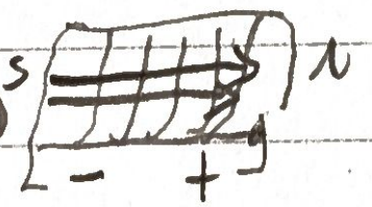
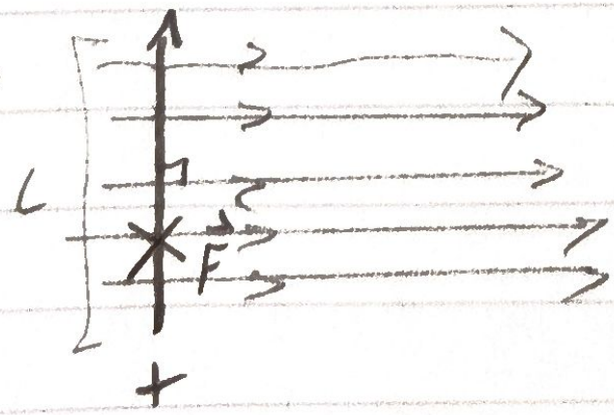
Oppg. 1 a)

ab) 

Oppg. 2 $F = \frac{k_0 I}{r}$ $\quad \odot \quad \Sigma F = \frac{k_0 \cdot 1}{r} - \frac{k_0 \cdot 2}{r} + \frac{k_0 \cdot 1}{r}$ $\quad$ 1 A

Oppg. 3 

$F = I \cdot L \cdot B$

$= 1{,}25\ A \cdot 9{,}60 \cdot 10^{-2}\ m \cdot 5{,}00 \cdot 10^{-3}\ T$

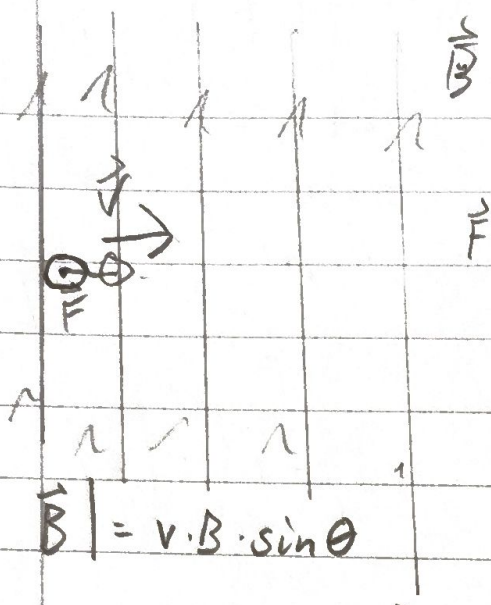
$= 0{,}0006\ N$

$\approx 6{,}00 \cdot 10^{-4}\ N$

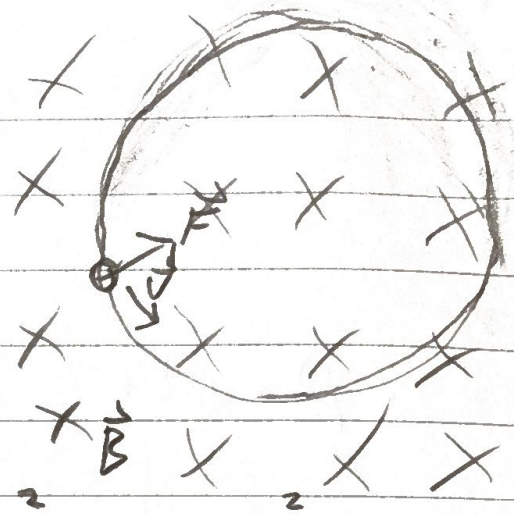
$\vec{F}$ går inn i papir-planet

BRUNNEN

oppg. 4



$\vec{F} = q\vec{v} \times \vec{B}$



$|\vec{v} \times \vec{B}| = v \cdot B \cdot \sin\theta$

$a = \frac{v^2}{r}$ $F = m\frac{v^2}{r}$

$m\frac{v^2}{r} = qvB$

$m\frac{v}{q \cdot B} = r$ $v = \frac{2\pi \cdot r}{T}$

$m\frac{2\pi r}{qBT} = r$

omrokering

$T = \frac{2\pi \cdot m}{q \cdot B}$

d) (1) $\Delta r < 0 \wedge \Delta T < 0$

(2) $\Delta r > 0 \wedge \Delta T = 0$

(3) $\Delta r < 0 \wedge \Delta T < 0$

08:30 30 minutt så langt

oppg. 5 $F_B = I \cdot L \cdot B$ rasten i ~~GG~~

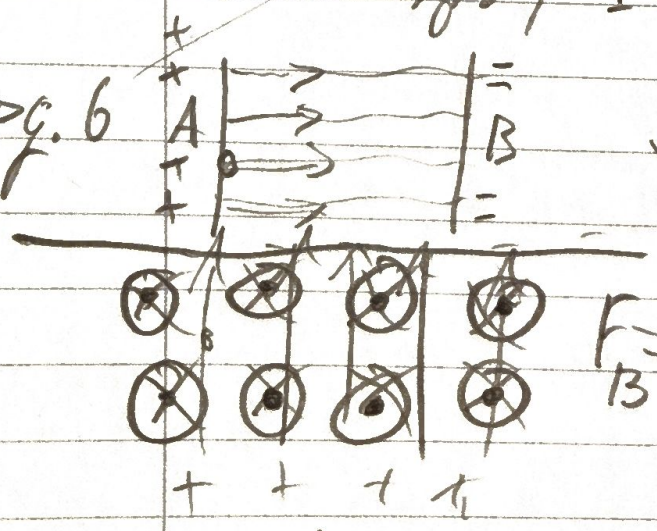
$K(G) = mg$ $K(I) = F_B + mg = I \cdot L \cdot B$

b) $0{,}072\,kg = m$ $B = 13\ T$

Glømte vindingar

c) mot høgre, $I = 0{,}88\ A$

oppg. 6



$E = \frac{U}{d}$ $F = \frac{U \cdot q}{d}$

$U = \frac{W}{Q}$

$E = U \cdot q = \frac{1}{2}mv^2$

$v = \sqrt{U \cdot q \cdot 2 \cdot \frac{1}{m}}$

$F_B = qvB = E_2 \cdot q$

$B = \frac{E_2}{v}$

Fasit antok $q = e$, elementærladningen.

KYS

$N = C \cdot m/s \cdot T$

kg m/s²

kg/sC